

### Objective

1. Manifest file
2. Deploy app using Deployment option
3. Display and Validate Deployment
4. Expose it to the external world
5. Test – POD Failures
6. Test – Scale up
7. Test – Scale down

Assumption:

- We are using 'NodePort'
-

### 1. Manifest file

#### a. Deployment-file

```
2. ---
3. apiVersion: apps/v1
4. kind: Deployment
5. metadata:
6.   name: deploy1
7. spec:
8.   replicas: 2
9.   selector:
10.    matchLabels:
11.      app: vclapp
12.   template:
13.     metadata:
14.       labels:
15.         app: vclapp
16.     spec:
17.       containers:
18.       - name: myapp
19.         image: nginx
20.
21.       ports:
22.       - containerPort: 80
```

Save the file as “deployment.yaml”

### a. service-port

```
1. ---
2. apiVersion: v1
3. kind: Service
4. metadata:
5.   name: node-port
6. spec:
7.   type: NodePort
8.   ports:
9.     - port: 88
10.       targetPort: 80
11.       nodePort: 30303
12.   selector:
13.     app: vclapp
```

Save the file as “service.yaml”

### 2. Folder structure looks like

```
● ubuntu@ip-172-31-24-80:~/projects/ex01$ tree
.
├── deployment.yaml
├── readme.md
└── service.yaml
```

### 3. Deploy app using Deployment

**\$ kubectl apply -f deployment.yaml**

```
● ubuntu@ip-172-31-24-80:~/projects/ex01$ kubectl get pods
No resources found in default namespace.
● ubuntu@ip-172-31-24-80:~/projects/ex01$ kubectl apply -f deploy.yaml
deployment.apps/deploy1 created
○ ubuntu@ip-172-31-24-80:~/projects/ex01$
```

**\$ kubectl get pods**

```
● ubuntu@ip-172-31-24-80:~/projects/ex01$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
deploy1-6f9dc7f445-lv89m            1/1     Running   0           86m
deploy1-6f9dc7f445-tlt7h            1/1     Running   0           86m
○ ubuntu@ip-172-31-24-80:~/projects/ex01$
```

### 4. Display and Validate deployment inside minikube

Login to the minikube Vm, and run

**projects/ex01\$ minikube ssh**

**\$ docker ps | grep nginx**

```
○ ubuntu@ip-172-31-24-80:~/projects/ex01$ minikube ssh
docker@minikube:~$ docker ps | grep nginx
3f8b6db503de   nginx          "/docker-entrypoint..." 16 hours ago    Up 16 hours
               k8s_myapp_deploy1-6f9dc7f445-tlt7h_default_752ea32b-3468-4805-b9de-cf4b8c7cb7df_0
dc42fabab9c1   nginx          "/docker-entrypoint..." 16 hours ago    Up 16 hours
               k8s_myapp_deploy1-6f9dc7f445-lv89m_default_f769b11d-ed1f-4015-b5f9-3e834604ccaa_0
docker@minikube:~$
```

Check each PODS

**Exit before next command**

**\$ kubectl describe pods/deploy1-6f9dc7f445-lv89m**

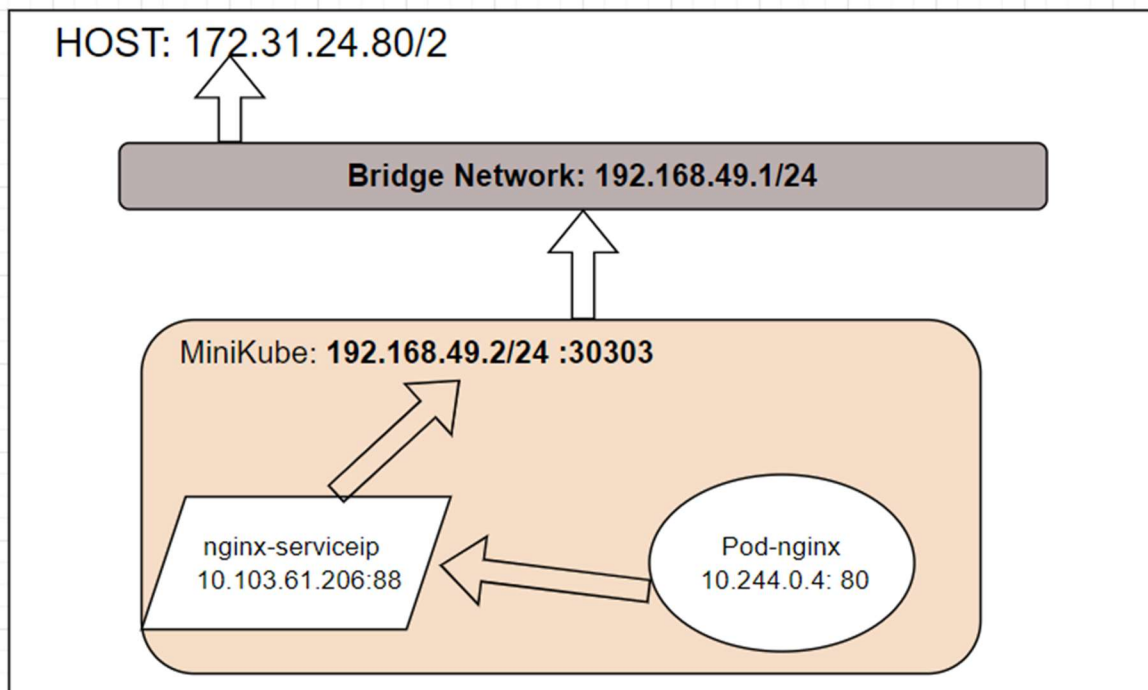
```
● ubuntu@ip-172-31-24-80:~/projects/ex01$ kubectl describe pods/deploy1-6f9dc7f445-lv89m
Name:          deploy1-6f9dc7f445-lv89m
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Tue, 24 Sep 2024 10:07:59 +0000
Labels:        app=vishwapp
               pod-template-hash=6f9dc7f445
Annotations:   <none>
Status:        Running
IP:            10.244.0.13
IPs:
  IP:          10.244.0.13
Controlled By: ReplicaSet/deploy1-6f9dc7f445
Containers:
  myapp:
    Container ID:  docker://dc42fabab9c11e55580bc9efb4be77def5127447abf908a377a9bec692067416
    Image:         nginx
    Image ID:      docker-pullable://nginx@sha256:04ba374043ccd2fc5c593885c0eacddebabd5ca375f9323666f28dfd5a9710e3
    Port:          80/TCP
    Host Port:     0/TCP
    State:         Running
      Started:     Tue, 24 Sep 2024 10:08:00 +0000
    Ready:         True
```

**\$ kubectl describe pods/deploy1-6f9dc7f445-tlt7h**

```
Events: <none>
• ubuntu@ip-172-31-24-80:~/projects/ex01$ kubectl describe pods/deploy1-6f9dc7f445-tlt7h
Name:          deploy1-6f9dc7f445-tlt7h
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Tue, 24 Sep 2024 10:07:59 +0000
Labels:        app=vishwapp
               pod-template-hash=6f9dc7f445
Annotations:   <none>
Status:        Running
IP:            10.244.0.14
IPs:
  IP:          10.244.0.14
Controlled By: ReplicaSet/deploy1-6f9dc7f445
Containers:
  myapp:
    Container ID:  docker://3f8b6db503def8615ae3f7cf0bc0481ea57a813332447da43d3492e8d1111801
    Image:         nginx
    Image ID:      docker-pullable://nginx@sha256:04ba374043ccd2fc5c593885c0eacddebabd5ca375f9323666f28dfd5a9710e3
    Port:          80/TCP
    Host Port:     0/TCP
    State:         Running
    Started:       Tue, 24 Sep 2024 10:08:00 +0000
```

### 5. Expose it to the external world ( Here the external world for the pod is minikube. Refe below diagram

**Public Ip for the Host : 4.3.2.1**



**\$ kubectl apply -f service.yaml**

```
ubuntu@ip-172-31-24-80:~/projects/ex01$ kubectl apply -f service.yaml
service/node-port created
ubuntu@ip-172-31-24-80:~/projects/ex01$ kubectl get service
NAME            TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes      ClusterIP     10.96.0.1     <none>         443/TCP          4d20h
node-port       NodePort      10.103.74.25  <none>         88:30303/TCP     7s
ubuntu@ip-172-31-24-80:~/projects/ex01$
```

Here the pod is exposed to external world on port “30303”

**\$ kubectl describe services/node-port**

```
ubuntu@ip-172-31-24-80:~/projects/ex01$ kubectl describe services/node-port
Name: node-port
Namespace: default
Labels: <none>
Annotations: <none>
Selector: app=vclapp
Type: NodePort
IP Family Policy: SingleStack
IP Families: IPv4
IP: 10.103.74.25
IPs: 10.103.74.25
Port: <unset> 88/TCP
TargetPort: 80/TCP
NodePort: <unset> 30303/TCP
Endpoints:
Session Affinity: None
External Traffic Policy: Cluster
Internal Traffic Policy: Cluster
Events: <none>
ubuntu@ip-172-31-24-80:~/projects/ex01$
```

This clearly shows that the service has created the LOADBALANCER.

The endpoint points to all the 3 PODS and  
The external port is working on “30303”